

Lecture 39-40: Primes, Euclidean algorithm, Congruences

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Outline

- **Trial division**
- The sieve of Eratosthenes
- Distribution of primes
- Euclidean algorithm
- Least common multipliers
- Congruences

Trial Division

Trial Division Theorem. If n is a composite positive integer, then n has a prime factor less than or equal to $\sqrt{n}$.

Proof.

Because n is a composite positive integer, there are $n_1, n_2 > 1$ such that $n = n_1 n_2$.

Then either $n_1 \leq \sqrt{n}$ or $n_2 \leq \sqrt{n}$.

For contradiction, suppose $n_1 > \sqrt{n}$ and $n_2 > \sqrt{n}$.

Then $n = n_1 n_2 > n$, a contradiction.

W.l.o.g. assume $n_1 \leq \sqrt{n}$.

According to the fundamental theorem of arithmetic, n_1 has at least one prime factor, say p . Evidently $p \leq n_1 \leq \sqrt{n}$.

Trial Division

Application of trial division theorem to primality test

To check that n is prime, it is sufficient to check
for every prime $p \leq \sqrt{n}$, p does not divide n .

Example. Show that 101 is a prime.

$$\lfloor \sqrt{101} \rfloor = 10.$$

All primes ≤ 10 are 2, 3, 5, 7.

$$101 \bmod 2 = 1, 101 \bmod 3 = 2,$$

$$101 \bmod 5 = 1, 101 \bmod 7 = 3.$$

Therefore for every $p = 2, 3, 5, 7$, p does not divide 101.

Thus 101 is a prime.

Trial Division

Application of trial division theorem to primality test

```
procedure trialDivisionPrimalityTest(n: positive integer > 1)
  ubd :=  $\lfloor \sqrt{n} \rfloor$     c := 2
  while c ≤ ubd
    if c | n then
      return false
    else
      c := c + 1
  return true. {return true iff n is a prime.}
```

Trial Division

Application of trial division theorem to primality test

procedure *trialDivisionPrimalityTest*(*n*: positive integer > 1)

ubd := $\lfloor \sqrt{n} \rfloor$ *c* := 2

while *c* ≤ *ubd*

if *c* | *n* **then**

return *false*

else

c := *c* + 1

return *true*. {return *true* iff *n* is a prime.}

$O(\sqrt{n} (\log n)^2)$ bit operations

Assuming the complexity of division
is $O(\log c \log n)$ bit operators

Trial Division

Application of trial division theorem to prime factorization.

Example. Compute the prime factorization for 7007.

$\lfloor \sqrt{7007} \rfloor = 83$. Starting from the prime 2.

None of 2, 3, 5 divides 7007. $7007 / 7 = 1001$.

$\lfloor \sqrt{1001} \rfloor = 31$. Starting from 7.

$1001 / 7 = 143$.

$\lfloor \sqrt{143} \rfloor = 11$. The prime numbers that are ≥ 7 and ≤ 11 are 7, 11.

7 does not divide 143 and $143 / 11 = 13$. 13 is a prime.

Thus $7007 = 7 \cdot 7 \cdot 11 \cdot 13 = 7^2 \cdot 11 \cdot 13$.

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The Sieve of Eratosthenes

Find all primes not exceeding a positive integer

All composites $n \leq 100$ satisfy $p \mid n$ for some $p = 2, 3, 5, 7$ ($\sqrt{100} = 10$)



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61	62	63	64	65	66	67	68	69	70
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81	82	83	84	85	86	87	88	89	90
91	92	93	94	95	96	97	98	99	100

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All $n: 2 < n \leq 100$
divisible by 2
are sieved out

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All $n: 3 < n \leq 100$
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The Sieve of Eratosthenes

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All $n: 5 < n \leq 100$
divisible by 5
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	11		13		17	19
			23			29
	31				37	
All $n: 5 < n \leq 100$ divisible by 5 are sieved out	41		43		47	49
			53			59
	61				67	
	71		73		77	79
			83			89
	91				97	

The Sieve of Eratosthenes

Find all primes not exceeding a positive integer

All composites $n \leq 100$ satisfy $p \mid n$ for some $p = 2, 3, 5, 7$ ($\sqrt{100} = 10$)

All $n: 7 < n \leq 100$
divisible by 7
are sieved out

1	2	3	4	5	6	7	8	9	10
11	12	13	14	15	16	17	18	19	20
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						97			

The Sieve of Eratosthenes

Theorem. There are infinitely many primes.

Proof.

The Sieve of Eratosthenes

Mersenne Primes

For almost 300 years, the largest known prime has been of special form $2^p - 1$, where p is a prime.

Examples.

The numbers

$$2^2 - 1 = 3, 2^3 - 1 = 7, 2^5 - 1 = 31 \text{ and } 2^7 - 1 = 127$$

are Mersenne primes, while

$$2^{11} - 1 = 2047 \text{ is not because } 2047 = 23 \cdot 89.$$

Marin Mersenne(1588–1648): French mathematician

The Sieve of Eratosthenes

Proposition. $2^n - 1$ is not a prime if $n > 1$ is not a prime.

Proof.

Outline

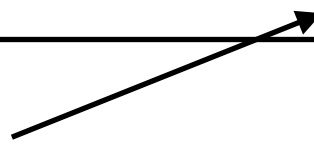
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Distribution of Primes

We have already known that there are infinitely many primes.
Given a positive integer n , **how many primes $\leq n$** are there ?

Prime Number Theorem (PNT). The prime counting function $\pi(n)$, which counts the number of primes not exceeding n , satisfies $\pi(n) \sim \frac{n}{\ln n}$, i.e. $\lim_{n \rightarrow \infty} \frac{\pi(n)}{n/\ln n} = 1$.

$\ln n = \log_e n$



Distribution of Primes

Prime Number Theorem

Conjectured by

Carl Friedrich Gauss and Adrien-Marie Legendre

First proved in 1896 independently by
Jacques Hadamard and Charles Jean de la Vallée Poussin

A consequence of PNT: $p_n \sim n \ln n$, where p_n is the n th prime.

Explicit bound for p_n (Rosser&Schoenfeld 1961, Dusart 1999)

For every $n \geq 6$, $n (\ln n + \ln \ln n - 1) < p_n < n (\ln n + \ln \ln n)$

Adrien-Marie Legendre (1752 – 1833): French mathematician

Jacques Hadamard (1865 – 1963): French mathematician

Charles-Jean de la Vallée Poussin (1866 – 1962): Belgian mathematician

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Euclidean Algorithm

Compute $\gcd(a, b)$ from prime factorizations of a and b ?

Prime factorizations of integers are difficult !!!

Efficient algorithm to compute $\gcd(a, b)$?

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Euclidean algorithm *The Elements*, Euclid

Proposition. Let $a, b \in \mathbb{Z}_+$ such that $a = bq + r$ for $0 \leq r < b$.

Then $\gcd(a, b) = \gcd(b, r)$.

Proof. It is sufficient to show $d \mid a$ and $d \mid b$ iff $d \mid b$ and $d \mid r$.

$d \mid a$ and $d \mid b \Rightarrow d \mid a$ and $d \mid (-bq) \Rightarrow d \mid (a - bq) = r$.

$d \mid b$ and $d \mid r \Rightarrow d \mid (bq)$ and $d \mid r \Rightarrow d \mid (bq + r) = a$.

Euclidean Algorithm

Example. $\gcd(287, 91)$

$$287 = 91 \cdot 3 + 14 \Rightarrow \gcd(287, 91) = \gcd(91, 14).$$

$$91 = 14 \cdot 6 + 7 \Rightarrow \gcd(91, 14) = \gcd(14, 7).$$

$$\gcd(14, 7) = 7.$$

Euclidean Algorithm

Example. Suppose $a \geq b$. Let $r_0 = a$ and $r_1 = b$.

$$r_0 = r_1 q_1 + r_2 \quad 0 \leq r_2 < r_1,$$

$$r_1 = r_2 q_2 + r_3 \quad 0 \leq r_3 < r_2,$$

$\cdot$
 $\cdot$
 $\cdot$

$$r_{n-2} = r_{n-1} q_{n-1} + r_n \quad 0 \leq r_n < r_{n-1},$$

$$r_{n-1} = r_n q_n.$$

$$\begin{aligned} \gcd(r_0, r_1) &= \gcd(r_1, r_2) = \gcd(r_2, r_3) = \dots \\ &= \gcd(r_{n-2}, r_{n-1}) = \gcd(r_{n-1}, r_n) = r_n. \end{aligned}$$

Euclidean Algorithm

procedure $gcd(a, b$: positive integers)

$x := a$

$y := b$

while $y \neq 0$

$r := x \bmod y$

$x := y$

$y := r$

return $x \{gcd(a, b) \text{ is } x\}$

Euclidean Algorithm

Assume $a \geq b$

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Always $x \geq y$ during the iteration of the while loop.

In two iterations of the while loop, $(a, b) \Rightarrow (b, c) \Rightarrow (c, c')$.

Then $a = bq + c$ and $b = cq' + c'$ for some $q, q' \geq 1$.

Therefore, $b = cq' + c' \geq c + c' > c' + c' = 2c'$, so $c' < b/2$.

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In **two** iterations, the value of y is **less than 1/2** of the original value.

The while loop iterates for at most $2 \lceil \log b \rceil$ times,
each iteration takes $O(\lceil \log a \rceil \lceil \log b \rceil)$ bit operations.

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$O(\log a (\log b)^2)$ bit operations

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Euclidean Algorithm

Bézout's theorem tells us $\gcd(a, b) = sa + tb$ for some s, t .

How to find s and t ?

Example. $\gcd(287, 91) = 7$

$$287 = 91 \cdot 3 + 14$$

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$$14 = 7 \cdot 2.$$

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$$14 = 7 \cdot 2.$$

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$$7 = 91 - (287 - 91 \cdot 3) \cdot 6 = (-6) \cdot 287 + 19 \cdot 91$$

Euclidean Algorithm

Bézout's theorem tells us $\gcd(a, b) = sa + tb$ for some s, t .

How to find s and t ?

$$\begin{array}{ll}
 r_0 = r_1 q_1 + r_2 & \longrightarrow r_2 = -r_1 q_1 + r_0 \\
 r_1 = r_2 q_2 + r_3 & \longrightarrow r_3 = -r_2 q_2 + r_1 \\
 \vdots & \vdots \\
 r_{n-3} = r_{n-2} q_{n-2} + r_{n-1} & \longrightarrow r_{n-1} = -r_{n-2} q_{n-2} + r_{n-3} \\
 r_{n-2} = r_{n-1} q_{n-1} + r_n & \longrightarrow r_n = -r_{n-1} q_{n-1} + r_{n-2} \\
 r_{n-1} = r_n q_n &
 \end{array}$$

$$r_n = -r_{n-1} q_{n-1} + r_{n-2} = -(-r_{n-2} q_{n-2} + r_{n-3}) q_{n-1} + r_{n-2} = \dots = t r_1 + s r_0.$$

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Least Common Multipliers

The *least common multiplier* of two positive integers a and b , denoted by $lcm(a, b)$, is the **smallest** positive integer divisible by both a and b .

Compute $lcm(a, b)$ from prime factorizations of a and b .

Suppose $a = p_1^{a_1} \cdot \dots \cdot p_n^{a_n}$ and $b = p_1^{b_1} \cdot \dots \cdot p_n^{b_n}$ where $a_1, \dots, a_n, b_1, \dots, b_n \geq 0$.

Then $lcm(a, b) = p_1^{\max(a_1, b_1)} \cdot \dots \cdot p_n^{\max(a_n, b_n)}$.

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Then $lcm(a, b) = p_1^{\max(a_1, b_1)} \cdot \dots \cdot p_n^{\max(a_n, b_n)}$.

$$gcd(a, b) = p_1^{\min(a_1, b_1)} \cdot \dots \cdot p_n^{\min(a_n, b_n)}$$

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$$gcd(a, b) = p_1^{\min(a_1, b_1)} \cdot \dots \cdot p_n^{\min(a_n, b_n)}$$

Examples. $lcm(180, 54) = lcm(2^2 3^2 5^1, 2^1 3^3 5^0) = 2^2 3^3 5^1 = 540$.

$gcd(180, 54) = gcd(2^2 3^2 5^1, 2^1 3^3 5^0) = 2^1 3^2 5^0 = 18$.

Least Common Multipliers

Proposition. Let a, b be positive integers. Then
$$ab = \gcd(a, b) \cdot \text{lcm}(a, b).$$

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$\text{lcm}(a, b) = p_1^{\max(a_1, b_1)} \cdot \dots \cdot p_n^{\max(a_n, b_n)}$

$\min(x, y) + \max(x, y) = x + y$ for every $x, y \in \mathbb{Z}$.

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$\text{lcm}(a, b) = p_1^{\max(a_1, b_1)} \cdot \dots \cdot p_n^{\max(a_n, b_n)}$

$\min(x, y) + \max(x, y) = x + y$ for every $x, y \in \mathbb{Z}$.

$$\begin{aligned} & \gcd(a, b) \cdot \text{lcm}(a, b) \\ &= p_1^{\min(a_1, b_1)} \cdot \dots \cdot p_n^{\min(a_n, b_n)} \cdot p_1^{\max(a_1, b_1)} \cdot \dots \cdot p_n^{\max(a_n, b_n)} \\ &= p_1^{\min(a_1, b_1) + \max(a_1, b_1)} \cdot \dots \cdot p_n^{\min(a_n, b_n) + \max(a_n, b_n)} \\ &= p_1^{a_1 + b_1} \cdot \dots \cdot p_n^{a_n + b_n} = ab. \end{aligned}$$

Outline

- Trial division
- The sieve of Eratosthenes
- Distribution of primes
- Euclidean algorithm
- Least common multipliers
- **Congruences**

Introduction

Congruence equations

“今有物不知其数，三三数之剩二（除以3余2），
五五数之剩三（除以5余3），
七七数之剩二（除以7余2），问物几何？”（孙子算经）

$$x \equiv 2 \pmod{3}$$

$$x \equiv 3 \pmod{5}$$

$$x \equiv 2 \pmod{7}$$

Introduction

What ?

Find solutions for congruence equations of the form

$$f(x) \equiv 0 \pmod{m}$$

$f: \mathbb{Z} \rightarrow \mathbb{Z}$ variable $m \in \mathbb{Z}_+$

Introduction

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Find solutions for congruence equations of the form

$$f(x) \equiv 0 \pmod{m}$$

$f: \mathbb{Z} \rightarrow \mathbb{Z}$ variable $m \in \mathbb{Z}_+$

Linear congruence equations $ax \equiv b \pmod{m}$

Introduction

Why ?

Make it **easier** to solve

certain otherwise very difficult problems about integers

Infinitely many cases $\Rightarrow$ Finitely many cases modulo m

E.g. Prove that $x^5 - x + 2 = 0$ has no integer solutions.

Introduction

Why ?

Make it **easier** to solve

certain otherwise very difficult problems about integers

Infinitely many cases $\Rightarrow$ Finitely many cases modulo m

E.g. Prove that $x^5 - x + 2 = 0$ has no integer solutions.

Many applications in math., computer science, ...:

e.g. Godel incompleteness theorems,

encryption algorithm RSA,

multiple range ambiguity resolution in many radar systems

Linear Congruences

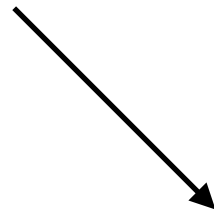
How to find all solutions of $ax \equiv b \pmod{m}$?

If there exists $\bar{a}$ such that $\bar{a}a \equiv 1 \pmod{m}$, then
 $\bar{a}ax \equiv \bar{a}b \pmod{m}$ and $x \equiv \bar{a}b \pmod{m}$.

Linear Congruences

How to find all solutions of $ax \equiv b \pmod{m}$?

$\bar{a}$: an **inverse** of a modulo m



If there exists $\bar{a}$ such that $\bar{a}a \equiv 1 \pmod{m}$, then

$\bar{a}ax \equiv \bar{a}b \pmod{m}$ and $x \equiv \bar{a}b \pmod{m}$.

Linear Congruences

Proposition. If a and m are relatively prime and $m > 1$,
then an inverse of a modulo m exists.
Moreover, this inverse is unique modulo m .

Examples.

Let $a = 2$ and $m = 3$.

$2a \equiv 1 \pmod{3} \Rightarrow 2$ is an inverse of 2 modulo 3.

Let $a = 2$ and $m = 4$.

$a'a \equiv 0 \text{ or } 2 \pmod{4}$ for every $a' \Rightarrow$

there does **not** exist an inverse of 2 modulo 4.

Linear Congruences

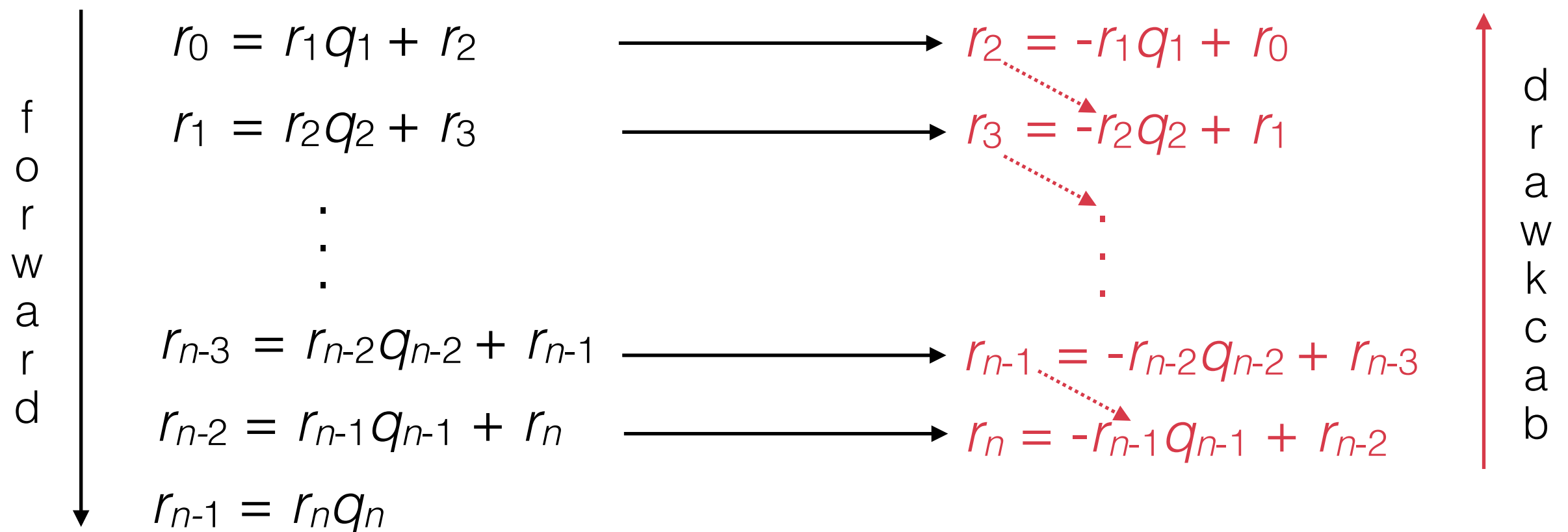
Proof.

Linear Congruences

Q: How to find Bézout coefficients ?

A: We can utilize Euclidean algorithm.

Two passes: Forward + Backward



$$r_n = -r_{n-1} q_{n-1} + r_{n-2} = -(-r_{n-2} q_{n-2} + r_{n-3}) q_{n-1} + r_{n-2} = \dots = t r_1 + s r_0.$$

Linear Congruences

Extended Euclidean algorithm

Find Bézout coefficients in **one pass**

Example. Let $a = 57$ and $m = 13$.

$$57 = 13 \cdot 4 + 5 \qquad 5 = 57 - 4 \cdot 13$$

Linear Congruences

Extended Euclidean algorithm

Find Bézout coefficients in **one pass**

Example. Let $a = 57$ and $m = 13$.

$$57 = 13 \cdot 4 + 5$$

$$5 = 57 - 4 \cdot 13$$

$$13 = 5 \cdot 2 + 3$$

$$3 = 13 - 2 \cdot 5 = (-2) \cdot 57 + 9 \cdot 13$$


Linear Congruences

Extended Euclidean algorithm

Find Bézout coefficients in **one pass**

Example. Let $a = 57$ and $m = 13$.

$$57 = 13 \cdot 4 + 5$$

$$13 = 5 \cdot 2 + 3$$

$$5 = 3 \cdot 1 + 2$$

$$5 = 57 - 4 \cdot 13$$

$$3 = 13 - 2 \cdot 5 = (-2) \cdot 57 + 9 \cdot 13$$

$$2 = 5 - 1 \cdot 3 = 3 \cdot 57 - 13 \cdot 13$$

Linear Congruences

Extended Euclidean algorithm

Find Bézout coefficients in **one pass**

Example. Let $a = 57$ and $m = 13$.

$$57 = 13 \cdot 4 + 5$$

$$13 = 5 \cdot 2 + 3$$

$$5 = 3 \cdot 1 + 2$$

$$3 = 2 \cdot 1 + 1$$

$$5 = 57 - 4 \cdot 13$$

$$3 = 13 - 2 \cdot 5 = (-2) \cdot 57 + 9 \cdot 13$$

$$2 = 5 - 1 \cdot 3 = 3 \cdot 57 - 13 \cdot 13$$

$$1 = 3 - 1 \cdot 2 = (-5) \cdot 57 + 22 \cdot 13$$

Linear Congruences

Extended Euclidean algorithm

Find Bézout coefficients in **one pass**

procedure *extendedEuclidean*(a, b : positive integers)

$x := a \quad y := b \quad s := 1 \quad t := 0 \quad s' := 0 \quad t' := 1$

while $y \neq 0$

$\{x = sa + tb, y = s'a + t'b\}$

$r := x \bmod y \quad q := x \operatorname{div} y \quad \{x = yq + r, \text{ thus } r = x - qy\}$

$s_1 := s - qs' \quad t_1 := t - qt' \quad \{r = s_1a + t_1b\}$

$x := y \quad y := r$

$s := s' \quad t := t' \quad s' := s_1 \quad t' := t_1$

return $(x, (s, t)) \quad \{x = \gcd(a, b) = sa + tb\}$

Linear Congruences

Extended Euclidean algorithm

Find Bézout coefficients in **one pass**

procedure *extendedEuclidean*(a, b : positive integers)

$x := a \quad y := b \quad s := 1 \quad t := 0 \quad s' := 0 \quad t' := 1$

while $y \neq 0$

$O((\log a \log b)^2)$ bit operations

$\{x = sa + tb, y = s'a + t'b\}$

$r := x \bmod y \quad q := x \operatorname{div} y \quad \{x = yq + r, \text{ thus } r = x - qy\}$

$s_1 := s - qs' \quad t_1 := t - qt' \quad \{r = s_1a + t_1b\}$

$x := y \quad y := r$

$s := s' \quad t := t' \quad s' := s_1 \quad t' := t_1$

return $(x, (s, t)) \quad \{x = \gcd(a, b) = sa + tb\}$

Linear Congruences

Solve $ax \equiv b \pmod{m}$ for relatively prime a and m

Let $\bar{a}$ be the inverse of a modulo m . Then

$$x \equiv \bar{a}b \pmod{m}.$$

Example. Solve $101x \equiv 18 \pmod{4620}$.

1601 is an inverse of 101 modulo 4620 $\Rightarrow$

$$(1601 \cdot 101)x \equiv (1601 \cdot 18) \pmod{4620} \Rightarrow$$

$$x \equiv 1098 \pmod{4620}$$

Linear Congruences

Solve $ax \equiv b \pmod{m}$ for **not** relatively prime a and m

Theorem.

Suppose $d = \gcd(a, m) > 1$, $a = da'$, and $m = dm'$.

Then $ax \equiv b \pmod{m}$ has a solution iff $d \mid b$.

Moreover, if $d \mid b$, then $ax \equiv b \pmod{m}$ iff $a'x \equiv b' \pmod{m'}$.

Linear Congruences

Proof of the theorem.

Linear Congruences

Proof of the theorem (continued).

Linear Congruences

Solve $ax \equiv b \pmod{m}$ for **not** relatively prime a and m

Example. Solve $8x \equiv 6 \pmod{18}$.

Recap

- Trial division

Trial Division Theorem. If n is a composite integer, then n has a prime divisor less than or equal to $\sqrt{n}$.

- The sieve of Eratosthenes

procedure *trialDivisionPrimalityTest*(n : positive integer > 1)

$ubd := \lfloor \sqrt{n} \rfloor$ $c := 2$

while $c \leq ubd$

if $c \mid n$ **then**

return *false*

else

$c := c + 1$

return *true*. {return *true* iff n is a prime.}

Trial division can also be used
for prime factorization

1	2	3		5		7	
11		13				17	19
		23					29
31						37	
41		43				47	
		53					59
61						67	
71		73					79
		83					89
						97	

Find all primes not exceeding a positive integer

Mersenne Primes

For almost 300 years, largest prime known has been of special form $2^p - 1$, where p is a prime.

Recap

Theorem. There are infinitely many primes.

- Distribution of primes

$$\text{lcm}(a, b) = p_1^{\max(a_1, b_1)} \cdot \dots \cdot p_n^{\max(a_n, b_n)}$$

- Least common multipliers

$$\text{gcd}(a, b) = p_1^{\min(a_1, b_1)} \cdot \dots \cdot p_n^{\min(a_n, b_n)}$$

- Euclidean algorithm

Proposition. Let $a, b \in \mathbb{Z}^+$ such that $a = bq + r$ for $0 \leq r < b$.
Then $\text{gcd}(a, b) = \text{gcd}(b, r)$.

procedure $\text{gcd}(a, b: \text{positive integers})$

$x := a$

$y := b$

while $y \neq 0$

$r := x \bmod y$

$x := y$

$y := r$

return $x \{ \text{gcd}(a, b) \text{ is } x \}$

$O(\log a (\log b)^2)$

Prime Number Theorem (PNT). The prime counting function $\pi(n)$, which counts the number of primes not exceeding n , satisfies $\pi(n) \sim \frac{n}{\ln n}$, i.e. $\lim_{n \rightarrow \infty} \frac{\pi(n)}{n/\ln n} = 1$.

A consequence of PNT: $p_n \sim n \ln n$, where p_n is the n th prime

Explicit bound for p_n (Rosser&Schoenfeld 1961, Dusart 1999)

For every $n \geq 6$, $n (\ln n + \ln n \ln n - 1) < p_n < n (\ln n + \ln n \ln n)$

Proposition. Let a, b be positive integers. Then

$$ab = \text{gcd}(a, b) \cdot \text{lcm}(a, b).$$

Can be used to describe $\text{gcd}(a, b)$ as linear combination of a and b .

$$\begin{array}{ll} r_0 = r_1 q_1 + r_2 & \longrightarrow r_2 = -r_1 q_1 + r_0 \\ r_1 = r_2 q_2 + r_3 & \longrightarrow r_3 = -r_2 q_2 + r_1 \\ \vdots & \vdots \\ r_{n-3} = r_{n-2} q_{n-2} + r_{n-1} & \longrightarrow r_{n-1} = -r_{n-2} q_{n-2} + r_{n-3} \\ r_{n-2} = r_{n-1} q_{n-1} + r_n & \longrightarrow r_n = -r_{n-1} q_{n-1} + r_{n-2} \\ r_{n-1} = r_n q_n & \end{array}$$

$$r_n = -r_{n-1} q_{n-1} + r_{n-2} = -(-r_{n-2} q_{n-2} + r_{n-3}) q_{n-1} + r_{n-2} = \dots = t r_1 + s r_0.$$

Proposition. If a and m are relatively prime and $m > 1$,
then an inverse of a modulo m exists.
Moreover, this inverse is unique modulo m .

Outline

- Congruences

```
procedure extendedEuclidean( $a, b$ : positive integers)
 $x := a$   $y := b$   $s := 1$   $t := 0$   $s' := 0$   $t' := 1$ 
while  $y \neq 0$ 
     $\{x = sa + tb, y = s'a + t'b\}$ 
     $r := x \bmod y$   $q := x \div y$   $\{x = yq + r, \text{ thus } r = x - qy\}$ 
     $s_1 := s - qs'$   $t_1 := t - qt'$   $\{r = s_1a + t_1b\}$ 
     $x := y$   $y := r$ 
     $s := s'$   $t := t'$   $s' := s_1$   $t' := t_1$ 
return  $(x, (s, t))$   $\{x = \gcd(a, b) = sa + tb\}$ 
```

Theorem.

Suppose $d = \gcd(a, m) > 1$, $a = da'$, and $m = dm'$.

Then $ax \equiv b \pmod{m}$ has a solution iff $d \mid b$.

Moreover, if $d \mid b$, then $ax \equiv b \pmod{m}$ iff $a'x \equiv b' \pmod{m'}$.

Homework

1. Section 4.3.

- Write a pseudocode for the sieve of Eratosthenes.
- Page 289, Exercise 32(c),(f).
- Page 290, Exercise 40(e),(i).
- Page 290, Exercise 52.

2. Section 4.4.

- Page 301, Exercise 2,8.

作业和答疑

1. 交作业时间：奇数周周一上课前（两周交一次）
2. 交作业方式（二选一）：
 - ◆ 通过<https://mooc.ucas.edu.cn/portal>课程网站
 - ◆ 上课现场提交
3. 答疑：助教崔乐天、刘易铖

Next Lecture

Solving Congruences